

Let $A \in \mathbb{M}_n(\mathbb{C})$, and define its numerical range by

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

Then, for every $m \geq 1$ and every matrix-valued polynomial

$$F(z) = \sum_{k=0}^d F_k z^k, \quad F_k \in \mathbb{M}_m(\mathbb{C}),$$

one has

$$\|F(A)\| \leq 2 \max_{z \in W(A)} \|F(z)\|,$$

where

$$F(A) = \sum_{k=0}^d F_k \otimes A^k \in \mathbb{M}_m(\mathbb{C}) \otimes \mathbb{M}_n(\mathbb{C}).$$

Equivalently, $W(A)$ is a complete 2-spectral set for A , uniformly over all matrix sizes m .